

ST 127 - TECHNICAL APPLICATION DOCUMENT

Visual identification of the model



ST 127 unit - reference rendering

MODEL ST 127

Document purpose

Indicative air-jet distribution

Reference conditions

Supply airflow: 950 m³/h

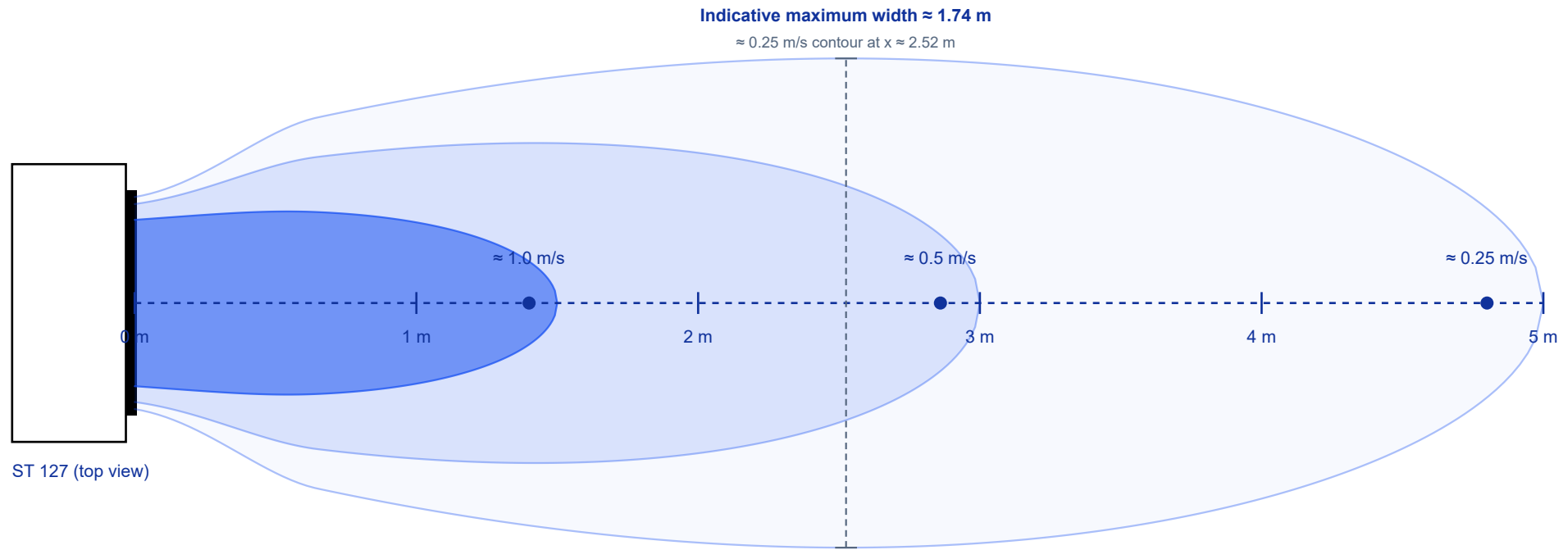
Studied room-height range: 2.60 to 3.00 m

Status

Indicative technical document - uncertified throws

Manufacturer test data or CFD validation required for final sizing

TOP VIEW - INDICATIVE AIR THROW DISTRIBUTION

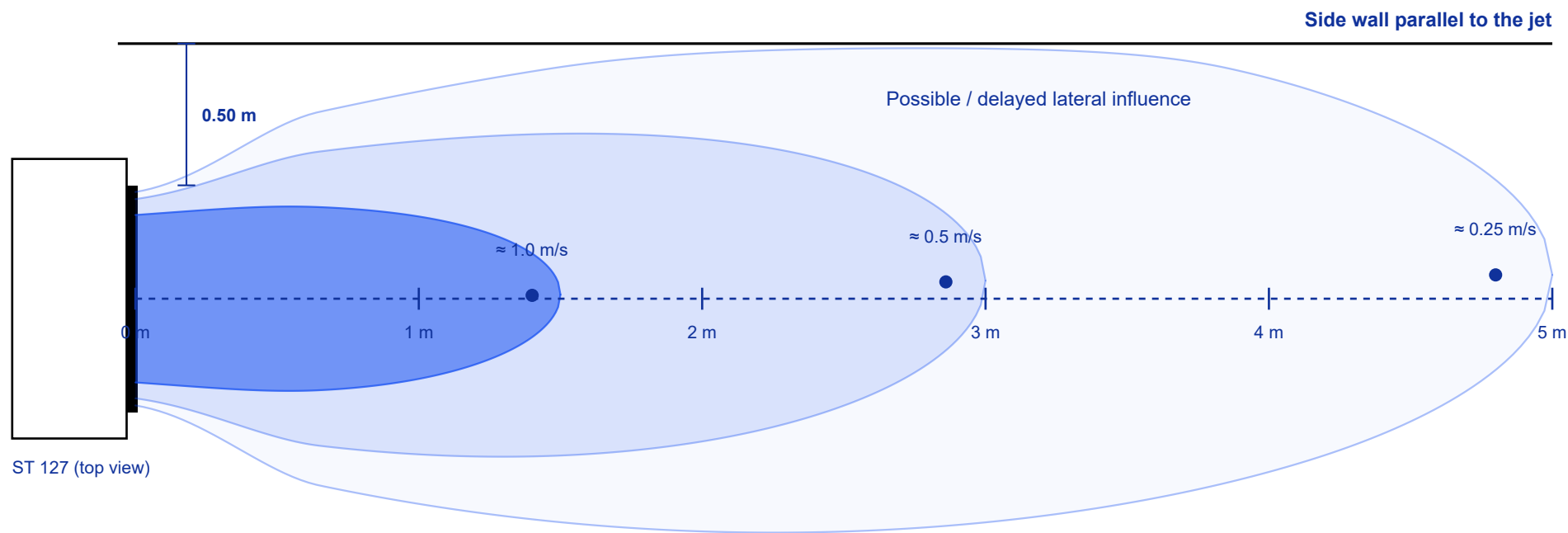
Airflow 950 m³/h

ST 127 (top view)

Horizontal projection · indicative parabolic-Gaussian model · opacity proportional to v^2 · uncertified throws

TOP VIEW - SIDE WALL AT 0.50 m

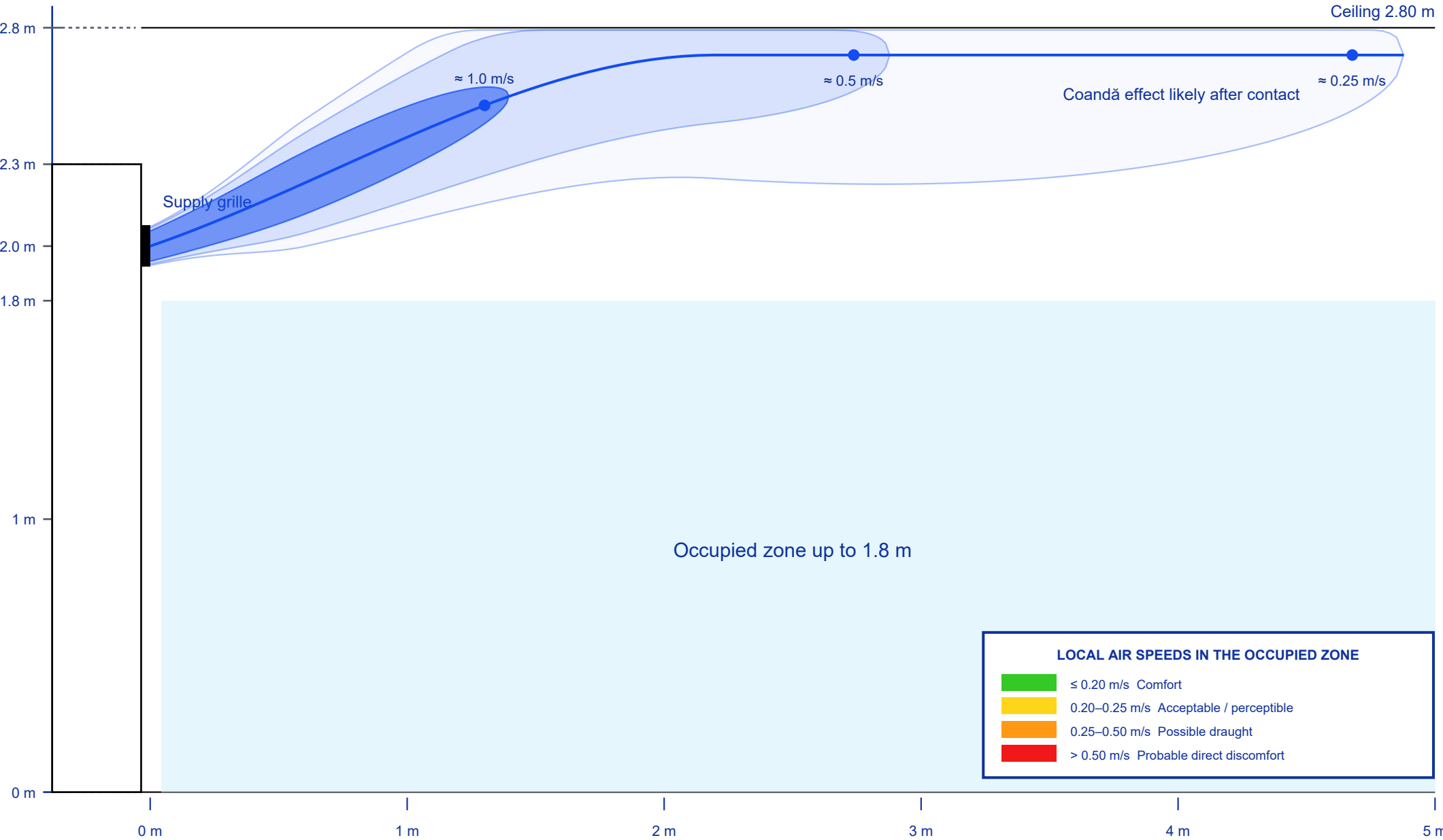
Indicative asymmetric distribution · airflow 950 m³/h



Horizontal projection · indicative Gaussian model · opacity proportional to v^2 · lateral index 42/100

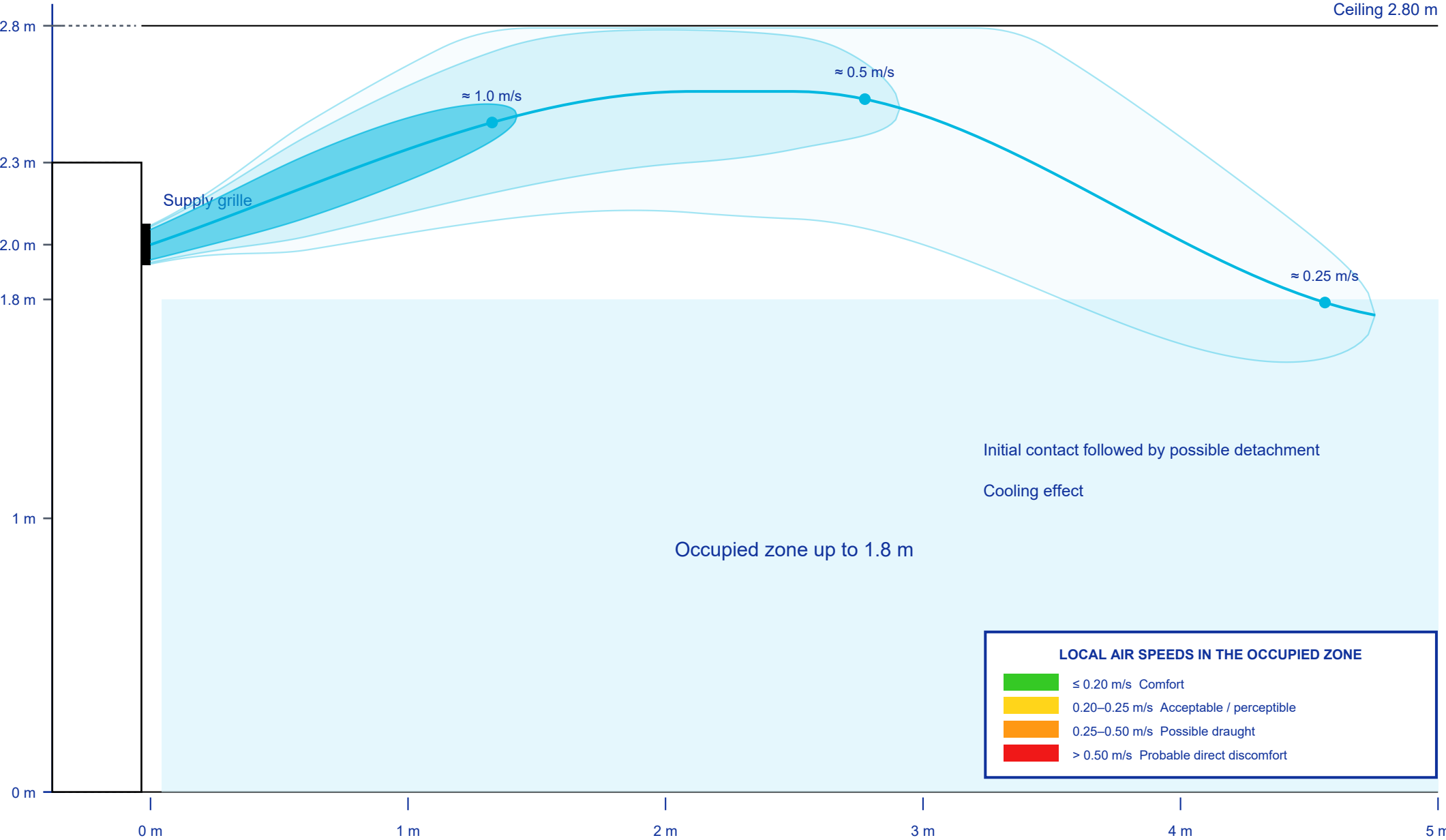
ISOTHERMAL CASE - INDICATIVE VERTICAL AIR THROW

Supply temperature $\approx$ room temperature · Airflow 950 m³/h



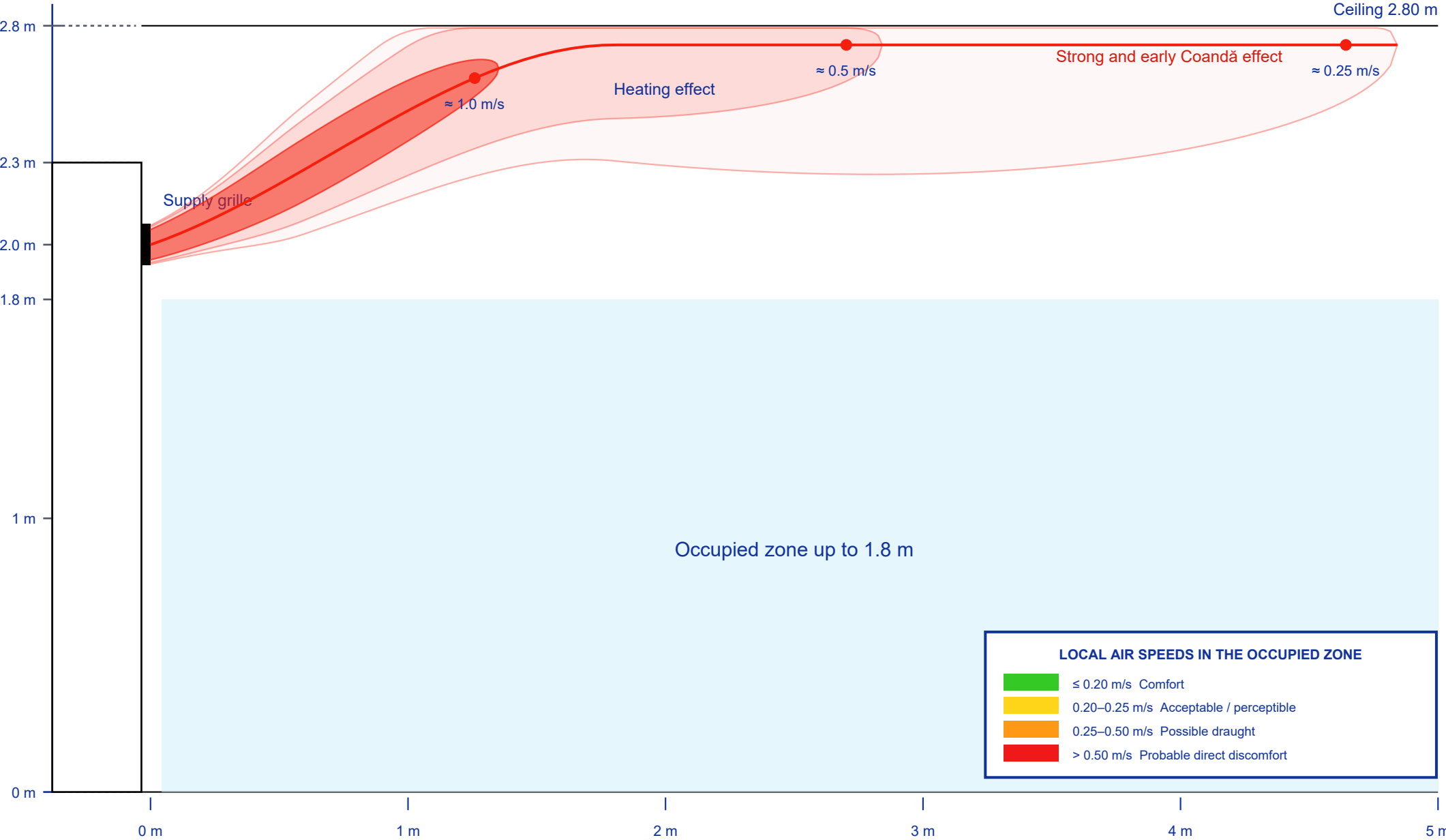
SUMMER CASE - INDICATIVE VERTICAL AIR THROW

Room 26 °C - supply 22 °C · Airflow 950 m³/h



WINTER CASE - INDICATIVE VERTICAL AIR THROW

Room 20 °C - supply 28 °C · Airflow 950 m³/h

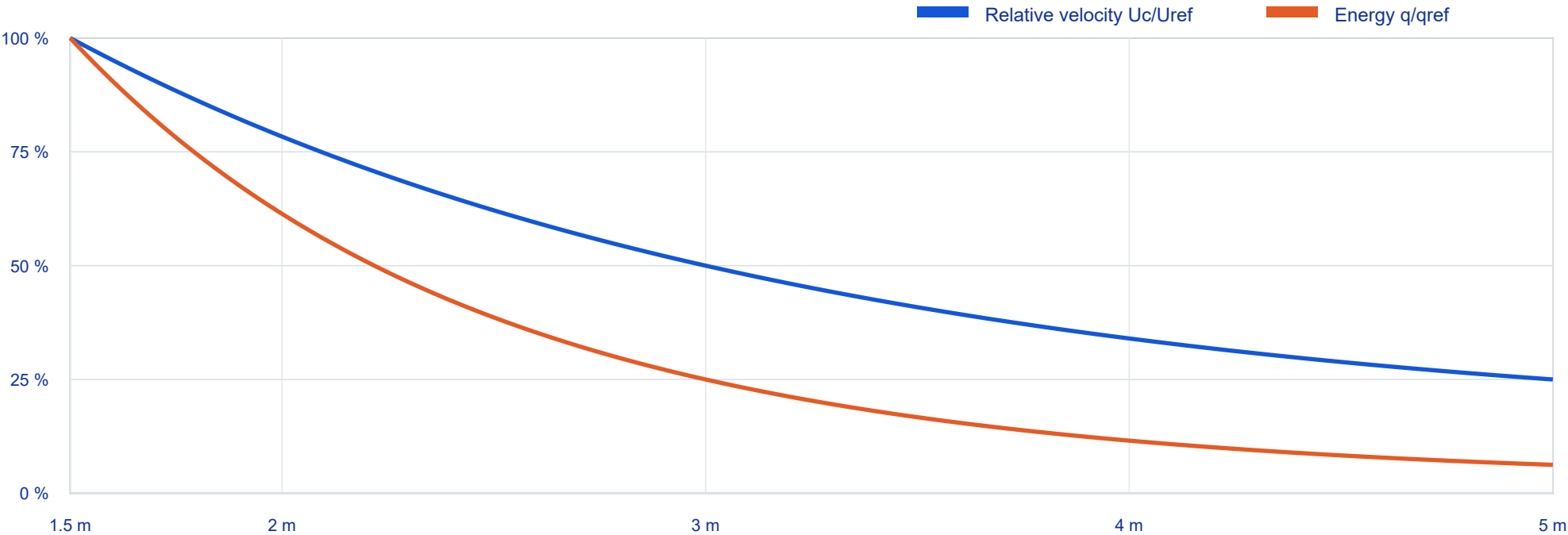


VELOCITY DECAY AND SQUARE LAW

Energy reference: 1.0 m/s at 1.5 m · air density 1.20 kg/m³

Dynamic pressure: $q(x) = \frac{1}{2} \rho U_c(x)^2$. For a local loss: $\Delta p = \zeta \cdot q$, with ζ to be determined by testing.

Comparison of relative velocity U_c/U_{ref} and relative energy $q/q_{ref} = (U_c/U_{ref})^2$.



Distance	Velocity U_c	Pressure q	Relative energy
1.5 m	1.00 m/s	0.6000 Pa	100.00 %
3.0 m	0.50 m/s	0.1500 Pa	25.00 %
5.0 m	0.25 m/s	0.0375 Pa	6.25 %

Graphic consequence: opacity of the 1.0 / 0.5 / 0.25 m/s zones = 100% / 25% / 6.25%.

The square law describes energy and losses through ζ ; jet spreading remains governed by turbulent mixing.

INTERACTION WITH THE CEILING

Supply axis at 2.00 m · airflow 950 m³/h

Studied room-height range: 2.60-3.00 m. This range does not limit product application.

Below 2.60 m, rooms are generally less representative of typical occupied spaces; a specific check remains possible.

Horizontal distances are calculated with the model's internal directional configuration; results are indicative and uncertified.

Ceiling height	Upper-edge interception	First contact of ≈ 0.25 m/s zone	Relative energy index
2.6 m	1.62 m	0.94 m	100 / 100
2.8 m	2.23 m	1.35 m	54 / 100
3.0 m	2.85 m	1.79 m	31 / 100

Indicative interpretation

- The energy index is proportional to Uc^2 at the geometric interception point and normalized to 100 at 2.60 m.
- It is neither an adhesion probability nor a normative Coandă-effect value.
- At 2.80 m, the upper edge reaches the ceiling at about 2.23 m; the ≈ 0.25 m/s zone may touch it at about 1.35 m.
- In summer, initial contact may be followed by detachment; in winter, buoyancy reinforces attachment.
- Heights outside 2.60-3.00 m remain possible but require a specific calculation or test.

RELATIVE COANDĂ-EFFECT SCALE - SIDE WALL

Wall parallel to the jet · airflow 950 m³/h

Distance measured perpendicular to the wall from the nearest edge of the supply opening.

Studied room-height range: 2.60 to 3.00 m; this range does not limit product application.

Relative geometric index: 100 = maximum predisposition. It is neither a probability nor a normative value.

Wall-to-outlet gap	Gap / outlet height	Lateral Coandă index	Indicative interpretation
0.10 m	0.7	100 / 100	Highly favoured
0.20 m	1.3	84 / 100	Highly favoured
0.30 m	2.0	66 / 100	Significant
0.40 m	2.7	53 / 100	Significant
0.50 m	3.3	42 / 100	Possible or delayed
0.60 m	4.0	33 / 100	Possible or delayed
0.80 m	5.3	21 / 100	Possible or delayed
1.00 m	6.7	13 / 100	Low

Requested case: wall-to-outlet gap of 0.50 m

- Indicative lateral geometric index $\approx 42/100$: possible or delayed influence.
- The wall should be parallel to the jet and continuous in the flow direction to favour attachment.
- Louvre orientation, turbulence, obstacles and buoyancy may increase or reduce this effect.
- Ceiling interaction may become dominant; ceiling and wall effects are not additive.
- This scale does not replace a certified outlet test curve or a CFD simulation.